



TUM SCHOOL OF COMPUTATION INFORMATION AND TECHNOLOGY

Professorship for Circuit Design

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Report PLL Lab

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1 Matlab Modelling

The closed loop transfer function in 1.1 has a bandwidth of 7.46MHz, and zero frequency about 1MHz. Which is enough for our input frequency range from 50M to 100MHz.

- $dV_p = 0.26 \cdot 5e7$
- $IntG = 2.857e6$
- $K_{prop} = K_{int} = 3.2e9$

Based on the values in MATLAB we adjust our parameters in cadence.

- $I_{cp} = 100\mu A$
- $K_{VCO} = 3.2G/V$
- Capacitors = 35pF, 3.5pF
- Resistor = 2.6KOhm

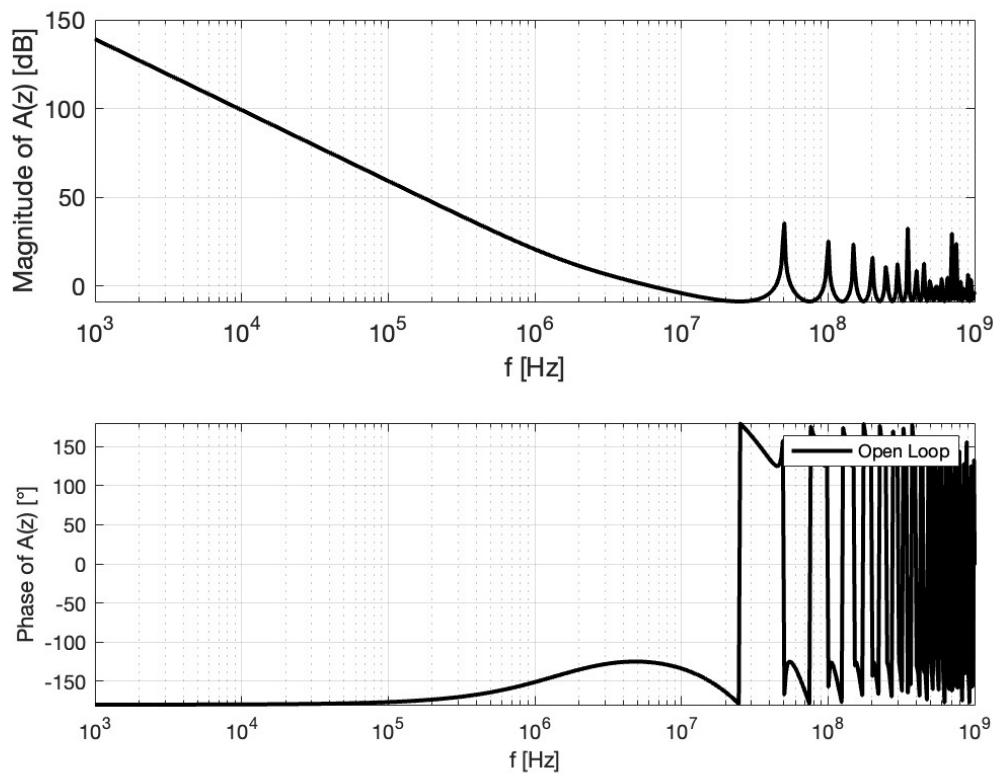


Figure 1.1: CL transfer function in MATLAB

2 Cadence Design Top-level

Figure 2.1 shows the top-level diagram of the charge pump-based phase-locked loop (CP-PLL) implemented in Cadence Virtuoso. The designed CP-PLL generates an output frequency that is 16 times the input reference frequency. This is achieved by minimizing the phase difference between the input clock and the divided version of the output clock. To ensure robust frequency locking and low-jitter performance, the design must minimize this phase difference effectively.

The operation of the CP-PLL is as follows: the generated output clock is divided by 16 and compared to the reference clock (REF_{clk}) using a phase-frequency detector (PFD). If the output clock (SYS_{clk}) leads the reference clock, the PFD asserts the DOWN signal. Conversely, if the reference clock leads, the UP signal is asserted. These signals drive the charge pump, which outputs a current I_{CP} depending on the pulse width. This current is then fed into the loop filter.

The loop filter consists of two paths: a proportional path designed using resistor R_c , and an integral path designed using capacitors C_c and C_f . The values of R_c and C_c were chosen based on MATLAB modeling of the PLL to meet the desired loop dynamics and stability. The capacitor C_f serves as a filter capacitor to smooth out rapid transitions in the control voltage to the VCO, which helps reduce output jitter.

To improve the jitter of the PLL, we use 13 voltage-controlled oscillators (VCOs) connected in parallel. These VCOs operate independently and their outputs are coherently combined. This parallel configuration helps to average out uncorrelated phase noise from individual VCOs, thereby improving the overall phase noise performance. Theoretically, this results in a reduction of output jitter by a factor of $\sqrt{13} \approx 3.6$, leading to a significantly cleaner output spectrum and enhanced performance of the CP-PLL.

Quantitatively, this technique yields a significant improvement:

- In dB: the phase-noise floor improves by $10 \log_{10} N$.
- For $N = 13$, this results in $10 \log_{10}(13) \approx 11.1$ dB improvement.
- In terms of RMS jitter: $\sigma_{\text{jitter,new}} = \frac{\sigma_{\text{single}}}{\sqrt{N}} = \frac{\sigma_{\text{single}}}{\sqrt{13}} \approx 0.28 \sigma_{\text{single}}$.

This reduction in jitter enhances the phase noise response of the VCO. The choice of the number of stages was made by pushing limits of the power boundary in the specifications.

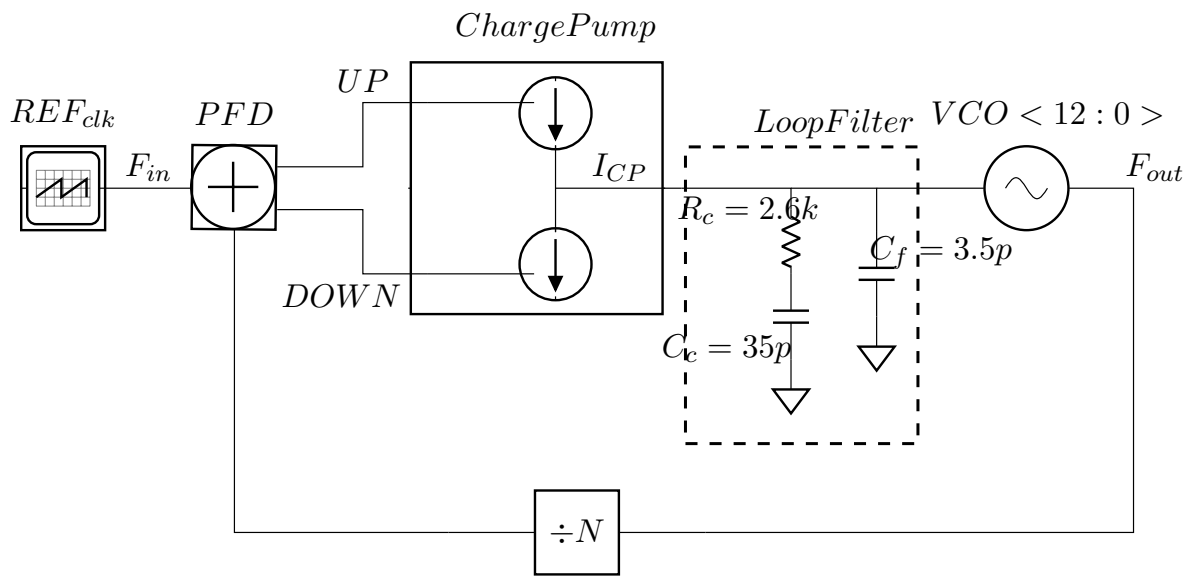


Figure 2.1: The diagram shows the top level structure of the charge pump based Phase Locked Loop.

3 Phase Frequency Detector

The target in the PFD's Design was to keep symmetry between UP/UPB and DOWN/DOWNB pulses in both delay and driving strength. To make sure both REF (Reference clock) and SYS (System clock) are driven with equal strength buffers, have been introduced at their respective terminals. The same logic was applied to the RST (Reset) signal. An additional buffer was added to create a delay, allowing UP and DOWN pulses to be active for a longer duration. The UPB and DOWNB signals, which control the reset signal, both have the same load capacitance. This was achieved by not using the UPB signal to control the charge pumps UP switch, but the UP signal instead. This, in turn, leads to the UP and DOWN signals driving the same load capacitance.

The flip-flops were designed based on a TSPC D-flip-flop with some modifications specific to the usecase in a PLL. The flip-flops for UP/UPB and DOWN/DOWNB pulse generation have the same size.

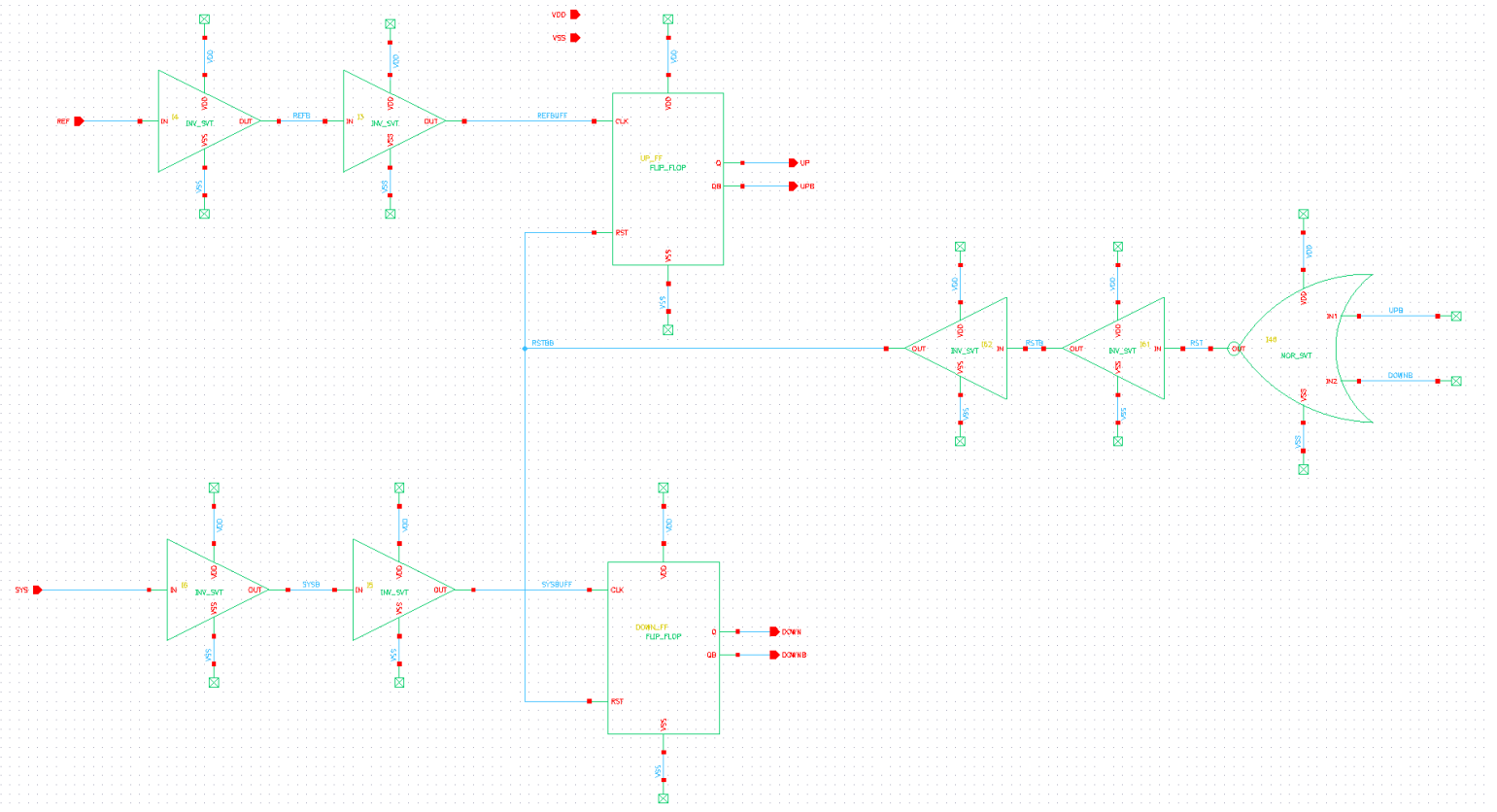


Figure 3.1: PFD schematic

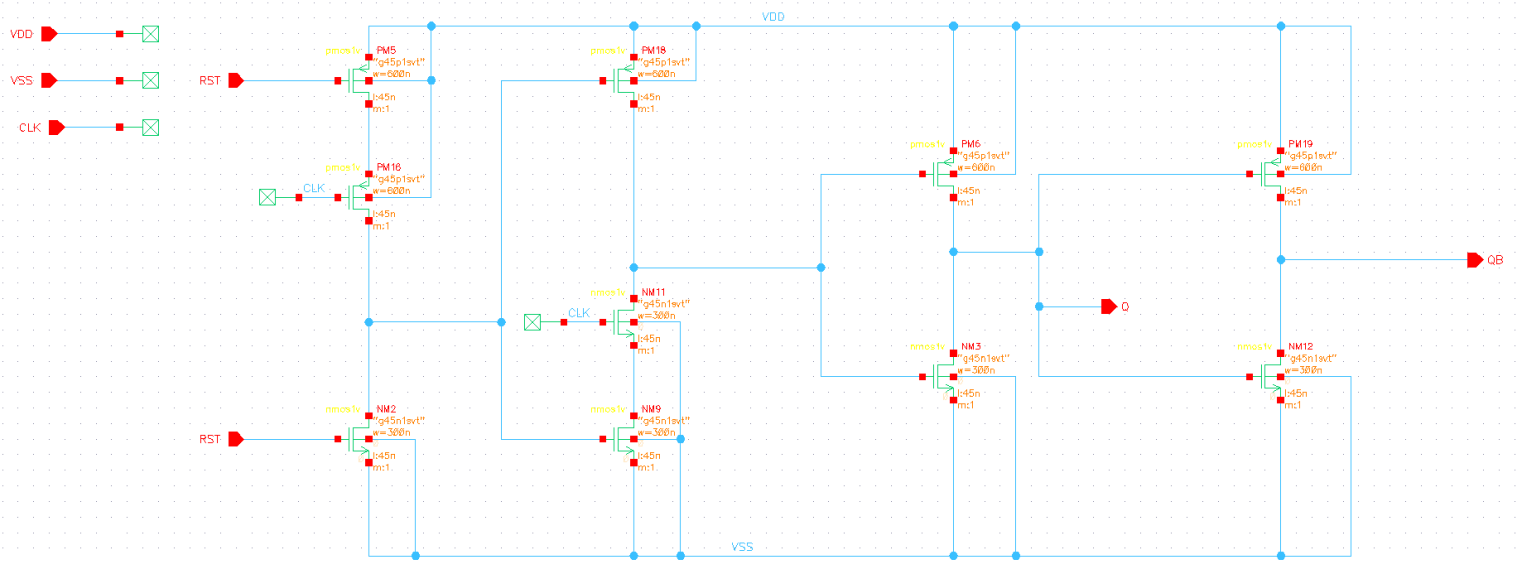


Figure 3.2: Flip-flop schematic

4 Charge Pump

The goal of the charge pump design was to provide similar current from both the UP and DOWN branches. The initial idea was to create a simple design and enhance the matching of the branches by adjusting the widths of the PMOS and NMOS current sources. The design can be seen in Figure 4.3. The problem with this approach was that while currents were well matched (about 1.7% mismatch) at 500mV, where tuning was done, mismatch increased across the output voltage range to up to 8% at 765mV (maximum VCO input voltage), see Figure 4.1. This mismatch can be explained by V_{DS} rising for the switching transistor, which ultimately leads to it falling out of saturation.

To combat this, the charge pump Design was changed to a self-biasing one inspired by [1]. The schematic can be seen in Figure 4.4. In the new design, multiple additions and changes were made. First replicas of the CP current sources and switches were introduced (PM17, PM16, NM18, NM20). These make it possible to bias the UP and DOWN current sources (PM12, NM12) independently. Two further devices were added (PM27 and NM27). These enable the better matching of this design. Since their Gates are controlled by the CP output voltage, they dynamically adjust the current flowing to/from the BIAS UP/BIAS DOWN nodes, thereby increasing/reducing bias voltage dynamically. This leads to the down path's current being reduced when the up path's current would naturally be reduced by rising output voltage. For the UP path, the same principle applies in reverse; it is being weakened by the feedback when CP OUT voltage is low. But since the VCO's tuning range only goes from 0.5V to 0.76V, this is not as important. The updated charge pumps mismatch ranges from 0% to 2% in this range, see Figure 4.2.

Also, with the redesign, the CP current has been increased (from around $58\mu A$ to around $100\mu A$), which made it possible to use a smaller resistance in the loop filter, leading to better noise performance but higher power consumption.

Another addition to the basic design used in the beginning where circuits to generate UPB and DOWN pulses inside the CP. These were added to keep the same load cap for all of the PFD's outputs and to give both pulses the same delay. The idea for this came from this source [2].

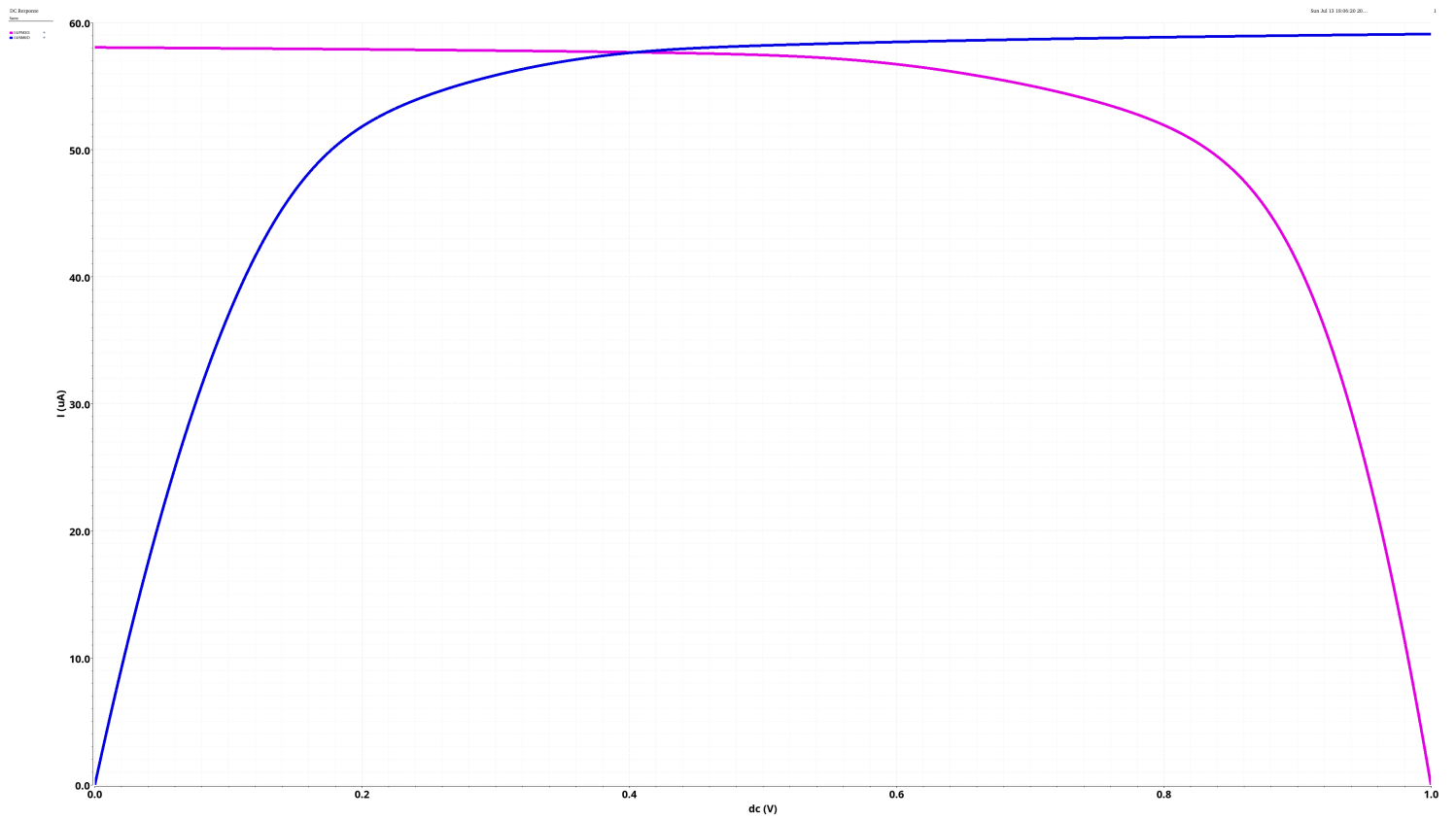


Figure 4.1: DC current behaviour of the uncompensated charge pump design for NMOS (blue) and PMOS (pink)

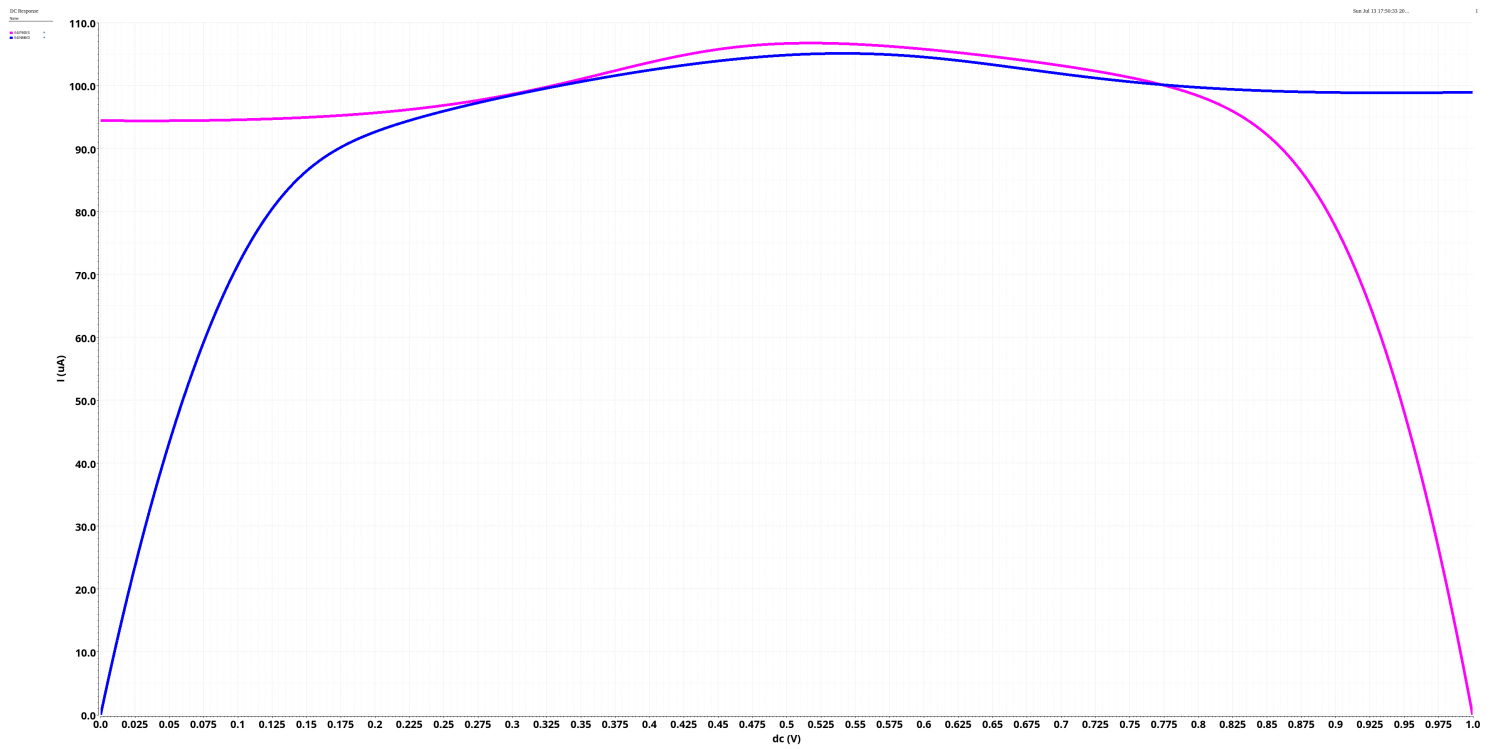


Figure 4.2: DC current behaviour of the compensated charge pump design for NMOS (blue) and PMOS (pink)

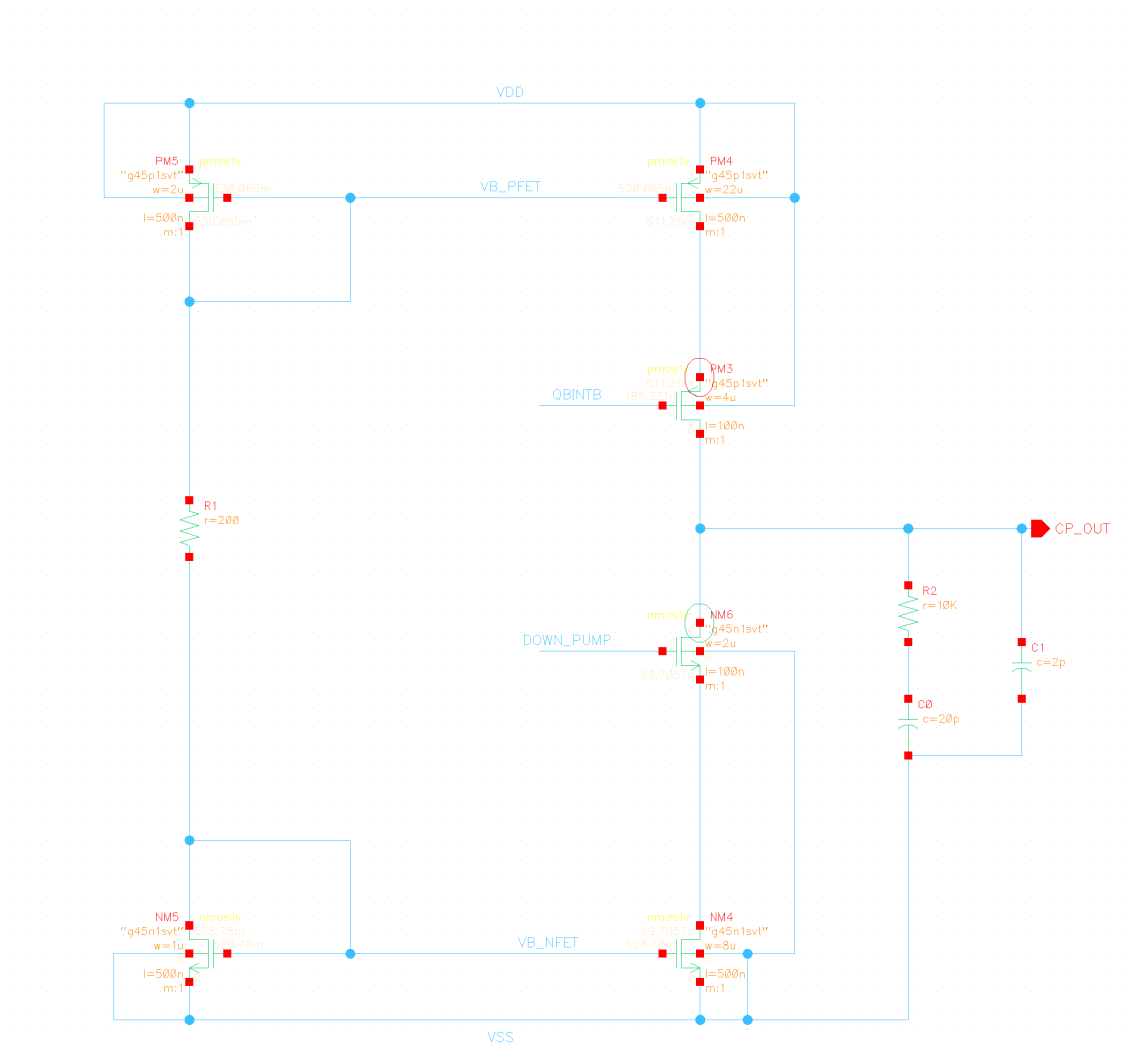


Figure 4.3: Design of the simpler charge pump used early in the project. PM4 is the pullup current source. PM3 acts as a switching device for the pullup. NM4 is the pulldown current source and NM6 acts as its current source.

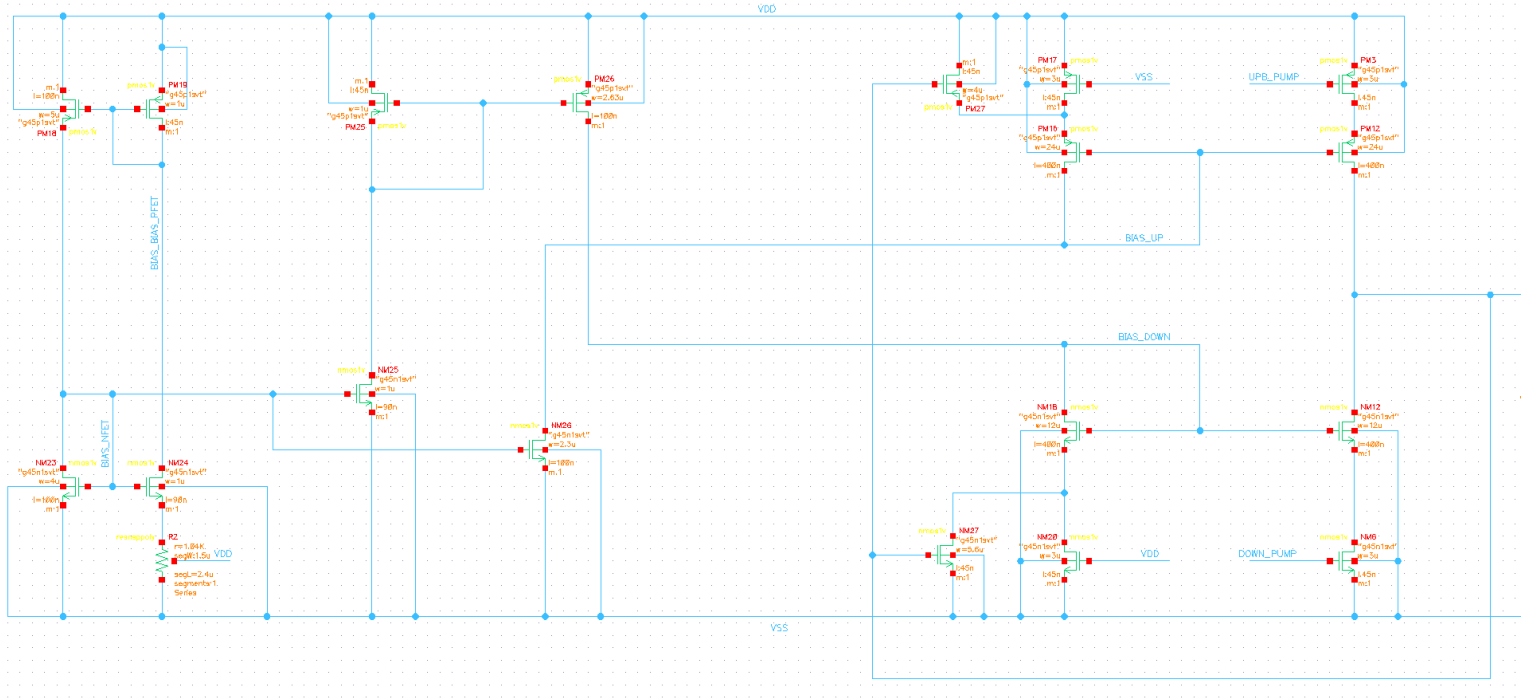


Figure 4.4: Schematic of the self compensating CP. The structure of the main current sources and switching devices remains (switching devices source now on VDD/VSS). Pullup and pulldown replika switches and current sources where introduced (PM17, PM16, NM18, NM20). Together with (PM27, NM27) they adjust the bias voltages dependent on the output voltage.

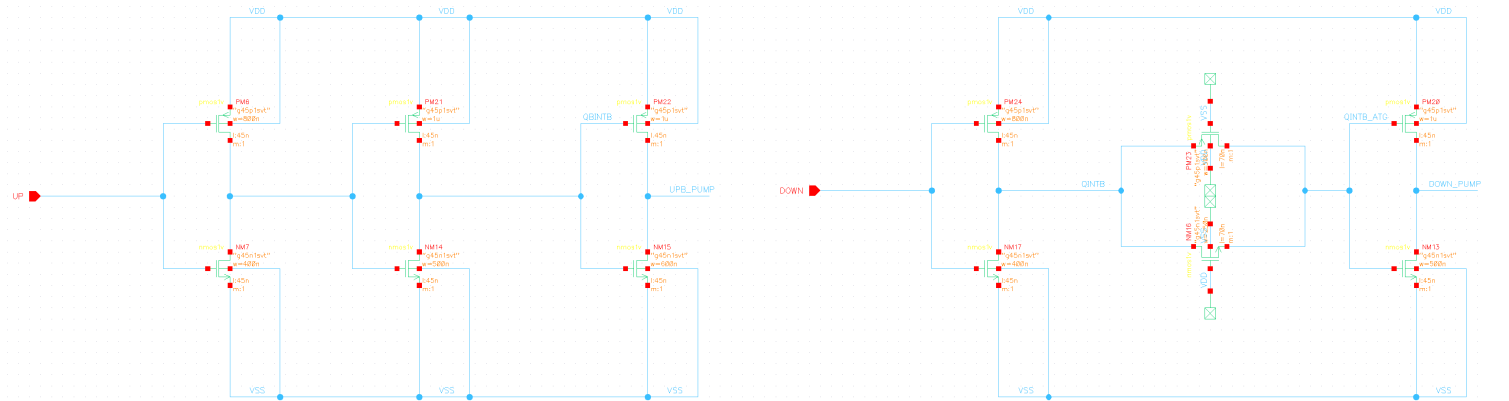


Figure 4.5: Schematic for the pulse drivers. On the left generation of the UPB PUMP pulse. On the right generation of the DOWN PUMP pulse with delay added by a transmission gate.

5 Loop Filter

The second order loop filter (see Figure 5.1) was designed based on the frequency acquired from the matlab tests. A range for the sizing of the components was found through the formulas from a guide by Renesas [3]. The loop bandwidth f_c was given as 7.5MHz. R_c (resistance) was calculated like this.

$$R_c = \frac{2 \cdot \pi \cdot f_c \cdot N}{I_{CP} \cdot K_{VCO}} = \frac{2 \cdot \pi \cdot 7.5MHz \cdot 16}{100\mu A \cdot 3.2\frac{GHz}{V}} = 2.35k\Omega \quad (5.1)$$

This value was later adjusted a little bit upwards to $2.6k\Omega$ to improve noise at higher frequencies. C_c (Compensation capacitance) was calculated with the following formula, where α was chosen between 3 and 5.

$$C_c = \frac{\alpha}{2\pi f_c R_c}$$

$$= \frac{\alpha}{2\pi \cdot 7.5MHz \cdot 2.6k\Omega} = \begin{cases} C_c(\alpha = 3) = \frac{3}{2\pi \cdot 7.5 \times 10^6 \cdot 2.6 \times 10^3} = 24.5pF \\ C_c(\alpha = 5) = \frac{5}{2\pi \cdot 7.5 \times 10^6 \cdot 2.6 \times 10^3} = 40.8pF. \end{cases} \quad (5.2)$$

In this range, multiple cap values were tested for noise performance, and it was ultimately decided to set $C_c = 35pF$. $C_f = 3.5$ was taken based on the 1/10 rule from the materials to this course.

The final step concerning the loop filter was to implement it into the design with components from the gpdk045. The sizes of the components can be seen in the schematics (link in appendix). It must be noted that the size of C_c ($l = 10\mu m, w = 270\mu m$, M2 in the schematic) is huge, since layout was no concern in this project, this can be tolerated. In a real scenario, the area available for C_c would of course have needed to be taken into consideration.

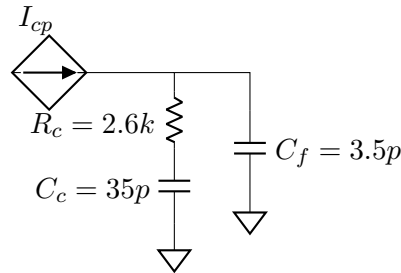


Figure 5.1: Schematic of the loop filter

6 Voltage Controlled Oscillator

6.1 is a 7 stage ring oscillator. The VCO needs to have a tuning range from 800MHz to 1.6GHz. In addition, the upper and lower MOSFETs allow the VCO to have a base frequency of about 600MHz, avoiding starting from 0; this helps our settling time. The output buffer ensures a perfect square wave at the output.

The challenge of VCO design is that it contributes the most to the jitter. Thus, we connected 13 VCOs in parallel to get the optimum jitter within our power budget.

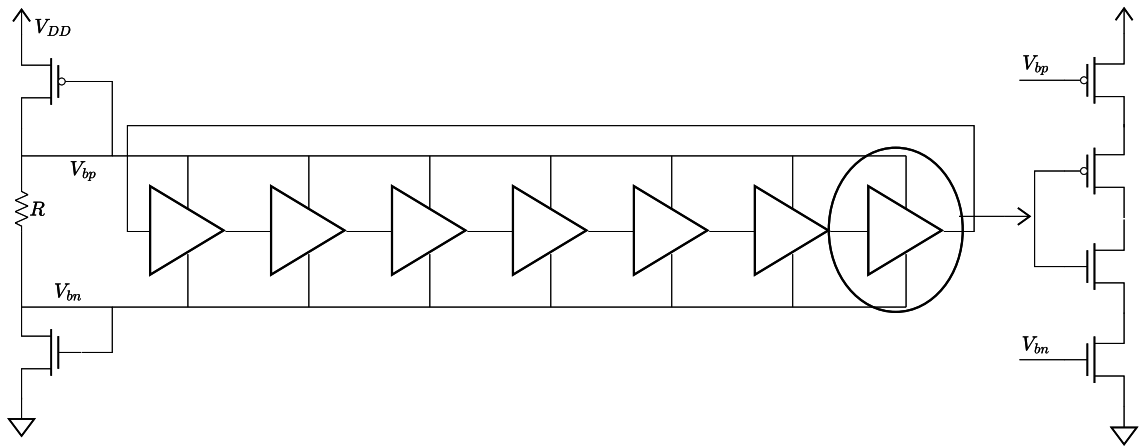


Figure 6.1: Schematic of the VCO

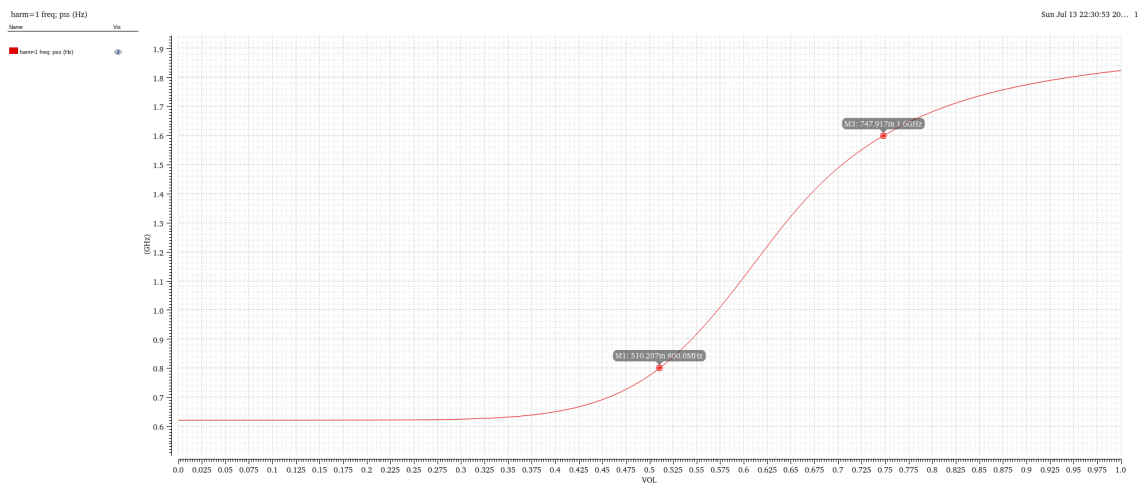


Figure 6.2: Tuning range of VCO

7 Divider

The divider is a simple digital frequency divider implemented using D flip-flops. To achieve an output frequency of 800 MHz from a 50 MHz reference, or 1.6 GHz from a 100 MHz reference, a constant divide ratio of 16 is employed. Figure 7.1 shows the structure of the divider. The D flip-flop used in this divider is a standard CMOS-based device with an asynchronous reset; its schematic is shown in Figure 7.2.

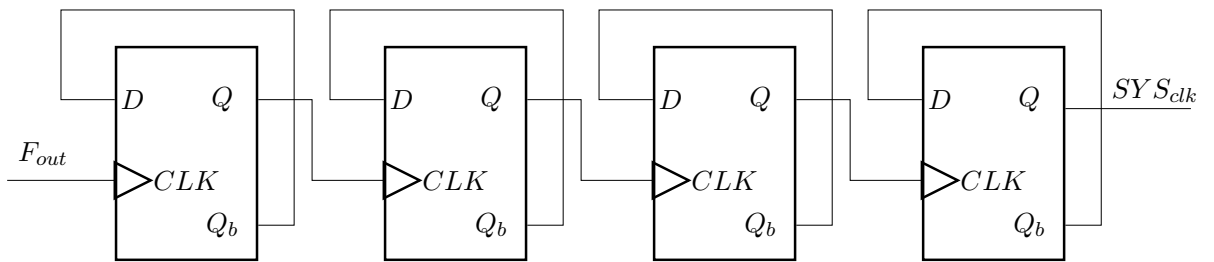


Figure 7.1: D-flipflop based divider. The divider divides F_{out} by a factor of 16 to generate $SY S_{clk}$

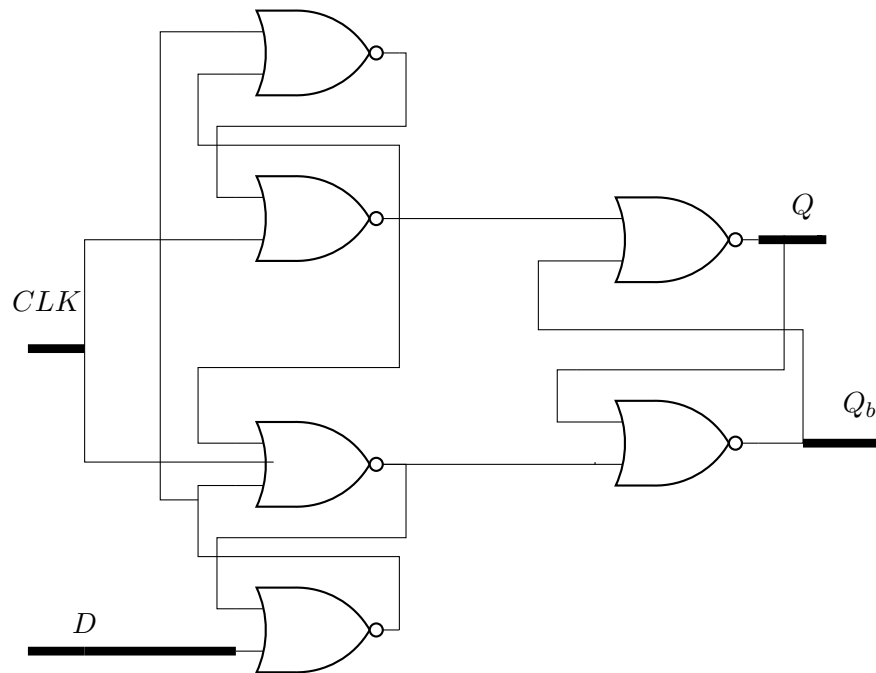


Figure 7.2: CMOS based D-flipflop divider

8 Simulation results

Tables 8.2 through 8.7 present the corresponding jitter and power metrics across various PVT corners for these input frequencies.

Figure 8.1 illustrates the settling time for the three input frequencies: 50MHz, 75MHz, and 100MHz.

Input clk	50MHz	75MHz	100MHz
Output clk	800MHz	1.2GHz	1.6GHz
Jitter	2.487ps	2.279ps	1.164ps
Power consumption	1.491mW	2.45mW	4.478mW
Settling time	650ns	680ns	500ns

Table 8.1: PLL Performance at Typical corner (@25°C)

	Nominal	FF 0°C	FF 80°C	FS 0°C	FS 80°C	SS 0°C
Jitter	2.453ps	5.49ps	1.91ps	5.105ps	2.864ps	7.09ps
Power	1.491mW	1.624mW	1.545mW	1.548mW	1.571mW	1.807mW

Table 8.2: PVT $f_{REF} = 50\text{MHz}$

	SS 80°C	SF 0°C	SF 80°C	TT 0°C	TT 80°C
Jitter	4.199ps	7.639ps	3.098ps	4.195ps	4.391ps
Power	1.652mW	1.419mW	1.449mW	1.498mW	1.52mW

Table 8.3: $f_{REF} = 50\text{MHz}$

	Nominal	FF 0°C	FF 80°C	FS 0°C	FS 80°C	SS 0°C
Jitter	2.293ps	2.271ps	1.903ps	2.123ps	2.029ps	2.292ps
Power	2.45mW	2.169mW	2.398mW	2.428mW	2.856mW	2.667mW

Table 8.4: $f_{REF} = 75\text{MHz}$

	SS 80°C	SF 0°C	SF 80°C	TT 0°C	TT 80°C
Jitter	3.397ps	2.602ps	2.658ps	2.232ps	2.086ps
Power	4.018mW	2.313mW	2.825mW	2.378mW	2.83mW

Table 8.5: $f_{REF} = 75\text{MHz}$

	Nominal	FF 0°C	FF 80°C	FS 0°C	FS 80°C	SS 0°C
Jitter	1.175ps	1.683ps	2.109ps	1.065ps	2.17ps	2.443ps
Power	4.478mW	3.193mW	4.915mW	3.792mW	5.394mW	5.237mW

Table 8.6: $f_{REF} = 100\text{MHz}$

	SS 80°C	SF 0°C	SF 80°C	TT 0°C	TT 80°C
Jitter	2.327ps	1.36ps	1.52ps	1.678ps	1.31ps
Power	4.708mW	3.781mW	5.128mW	3.763mW	5.22mW

Table 8.7: $f_{REF} = 100\text{MHz}$

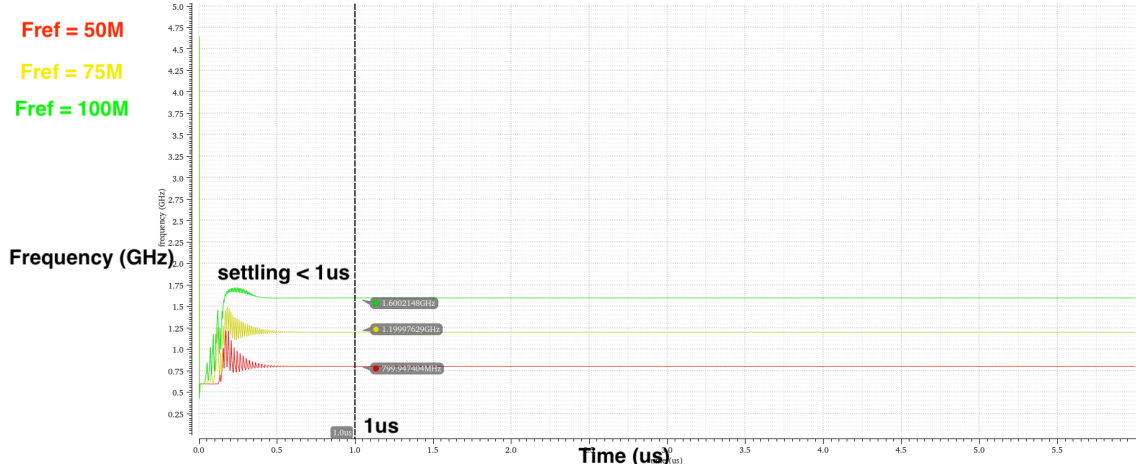


Figure 8.1: Settling time for 50MHz, 75MHz, 100MHz input frequency. Clearly in all the frequencies are settling under $1\mu\text{s}$.

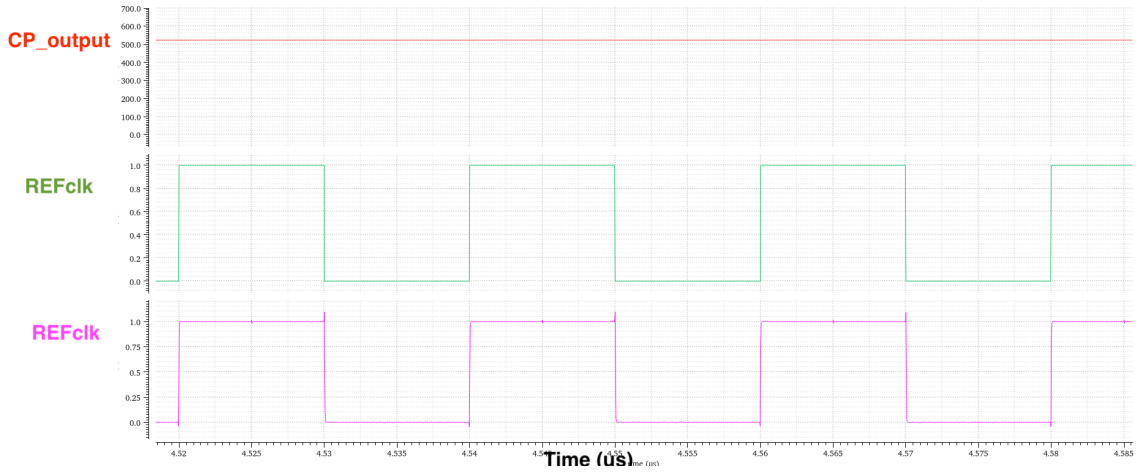


Figure 8.2: Comparison of REF_{clk} and SYS_{clk} at 50MHz. The PLL is locking at the given frequency and can be clearly visualized through the steady charge pump output.

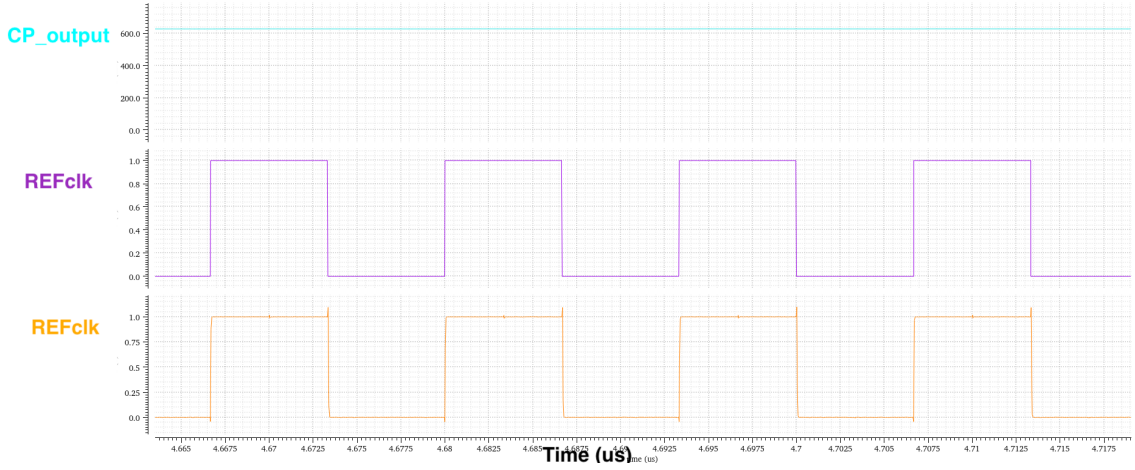


Figure 8.3: Comparison of REF_{clk} and SYS_{clk} at 75MHz. The PLL is locking at the given frequency and can be clearly visualized through the steady charge pump output.

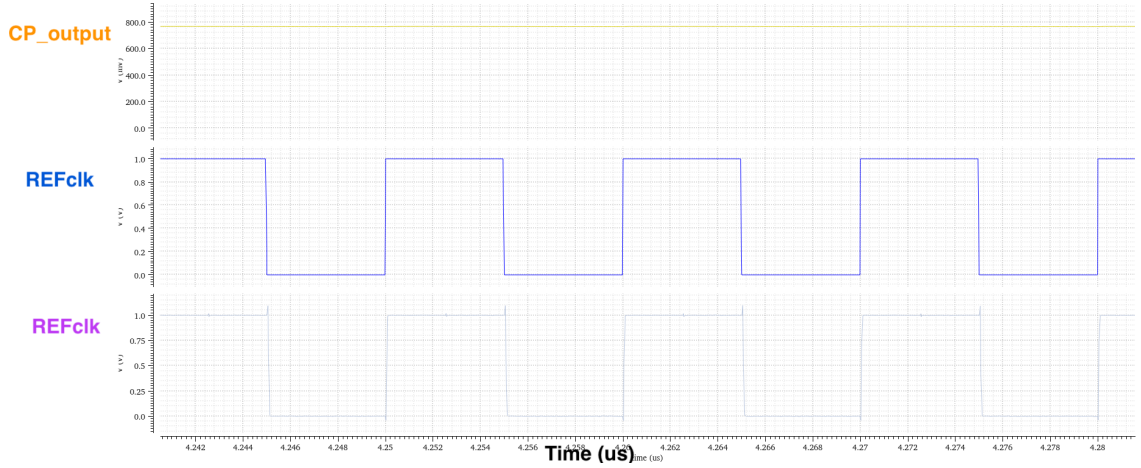


Figure 8.4: Comparison of REF_{clk} and SYS_{clk} at 100MHz. The PLL is locking at the given frequency and can be clearly visualized through the steady charge pump output.

To analyze the effect of mismatch and process variations on PLL performance, we performed a 50-point Monte Carlo simulation and characterized the resulting jitter. However, running a larger number of simulations (e.g., 1000) was computationally expensive. To estimate the expected behavior over a larger population, we applied bootstrapping on the 50 samples. This allowed us to extrapolate the mean jitter and estimate confidence intervals, providing a statistically reliable picture of performance with minimal simulation effort. The results of monte carlo simulations are shown in Fig. 8.5 and 8.6.

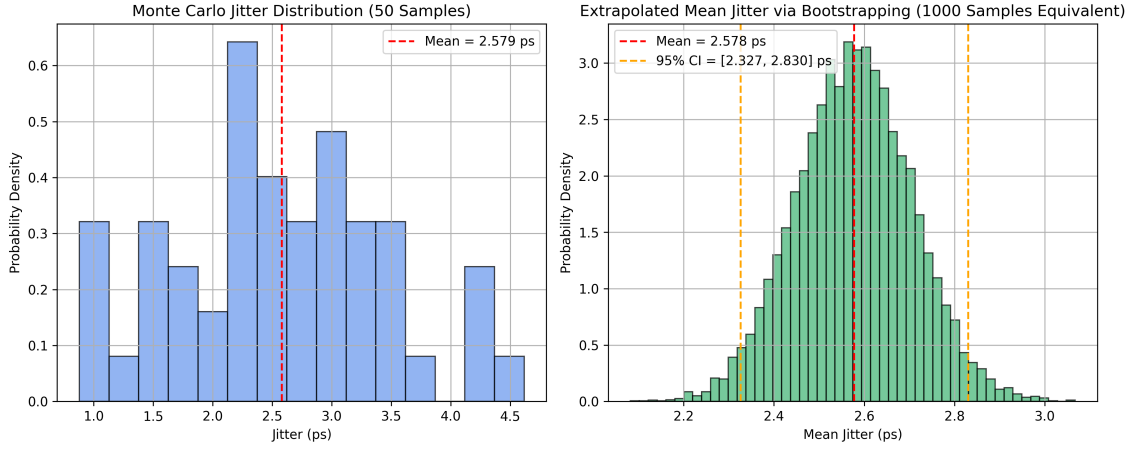


Figure 8.5: Probability distribution function of jitter of the PLL at $REF_{clk} = 50M$ from monte-carlo estimate. We use bootstrapping technique to extrapolate it for 1000 samples.

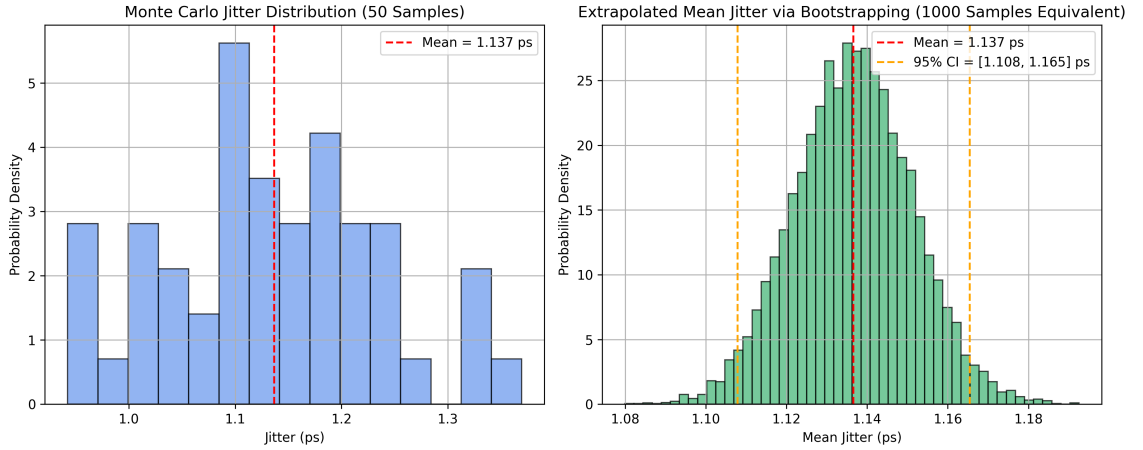


Figure 8.6: Probability distribution function of jitter of the PLL at $REF_{clk} = 100M$ from monte-carlo estimate. We use bootstrapping technique to extrapolate it for 1000 samples.

9 Appendix

Link to top-level schematic/maestro:

```
/nas/ei/share/TUEILSE/AC_Praktikum/sose25/ge89sec/cadence/  
YC_PLL/testbench_CLOSED_LOOP_final/schematic
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```
/nas/ei/share/TUEILSE/AC_Praktikum/sose25/ge89sec/cadence/  
YC_PLL/testbench_CLOSED_LOOP_final/maestro_1
```

Bibliography

- [1] T.-H. Lin, C.-L. Ti, and Y.-H. Liu, “Dynamic current-matching charge pump and gated-offset linearization technique for delta-sigma fractional- n plls,” *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 56, no. 5, pp. 877–885, 2009.
- [2] B. Razavi, *Design of CMOS Phase-Locked Loops: From Circuit Level to Architecture Level*. Cambridge University Press, 2020.
- [3] “link: <https://www.renesas.cn/zh/document/apn/pll-loop-filter-design-and-fine-tuning>,” 2024.